

The Amazing 'i'Phone

Apple's iPhone is a thoughtfully designed product that integrates best-of-breed technologies from Infineon, Samsung, Broadcom, Intel, Linear Technologies, STMicroelectronics, Triquint, NXP Semiconductors, Texas Instruments, Marvell and many more

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● Phone 3G, a marvel product from Apple, has enticed users with its applications and amazed techies with its inside. It is a global effort of more than 10,000 people from over 30 countries on three continents. Apple has created a difference with its most innovative design and, of course, the software that gives iPhone its unique identity.

As shown in Fig. 1, the macro-level teardown of an iPhone 3G has basically five elements—display glass, LCD, main logic board, battery and back panel. The closeup of the logic board in Fig. 2 shows all the ICs marked with details.

Touch screen and display

iPhone features a revolution in user interface: Its large multi-touch screen responds to both touch points and their movements (like spreading fingers apart to zoom in or pinching in to zoom out) simultaneously.

The screen has two main parts, the



Fig. 1: iPhone 3G taken apart

glass and the digitiser/LCD. iPhone 2G had these components glued together, which made replacement of screens (entire display units) very expensive. Good news is that the glass and the LCD screen are separate in 3G.

The capacitive touch-screens used in 3G come from Sharp and Sanyo Epson. Broadcom's BCM5974 chip is used as the touch controller and Texas Instruments (TI) provides the line driver. National Semiconduc-

tors' LM2512AA provides display interface. Two boards (PCBs)—logic and communication—found in the earlier iPhone are now integrated into a single board that is laid along the length.

Processor and memory

Samsung has provided the DDR SDRAM and the main processor (S3C6400 based on ARM1176 core). S3C6400 is a 16-/32-bit RISC micro-processor designed to provide cost-effective, low-power and high-performance application solutions for mobile phones. The same processor has been used in iPhone 2G too. Externally, the main memory comes in 8GB and 16GB NAND flash from Toshiba. SST's SST25VF080B chip provides 1MB serial flash.

Power management

Infineon's SMARTi power management IC SMP3i 6820 is optimised

What's new in iPhone 3GS

There is no significant difference between the logic boards of 3G and 3GS. The 'S' associated with the latter stands for speed. The main Samsung processor used in 2G and 3G has been replaced with S5PC100 in 3GS. It has been found that Samsung's fastest processor in 3GS is set to run at 667 MHz when the maximum can be set to 833 MHz. Apple is under-clocking the processor speed to about 72 per cent seemingly for longer battery life. A faster processor at lower clock frequency will consume less power than a slower processor. The chip also comes with HD video capability of 720p (720 vertical lines with progressive scanning). The memory has been doubled to 256 MB.

iPhone 3GS includes iPhone OS 3.0 with over 100 new features such as cut, copy and paste, MMS, spotlight search and a landscape keyboard. iPhone 3GS customers will get access to more than 50,000 applications from the App Store.